

Commerce XI Maths App

Exercise 4.2

Detailed Answers with Explanations and Steps

Mathematics & Statistics, Part 2, Unit 4
Bivariate Frequency Distribution and Chi Square Statistic
Total questions: 6

Question 1 - ENTER ANSWER · PG-52 · EX-Q1

Following table shows the classification of applications for secretarial and for sales positions according to gender. Calculate the value of χ^2 statistic.

	Offered	Denied
Male	75	150
Female	25	50

Correct answer: 0

Explanation and steps

Observed and expected frequencies are proportional, so $\chi^2 = 0$.

Question 2 - ENTER ANSWER · PG-53 · EX-Q2

200 teenagers were asked which take out food do they prefer – French fries, burger or pizza. The results were –

	French Fries	Burger	Pizza
Boys	6	20	24
Girls	18	40	92

Compute χ^2 statistic.

Correct answer: 3.37

Explanation and steps

Calculate expected frequencies from the margins and sum $(O - E)^2/E$ to obtain approximately 3.37.

Question 3 - ENTER ANSWER · PG-53 · EX-Q3

A sample of men and women who had passed their driving test either in 1st attempt or in 2nd attempt were surveyed. Compute χ^2 statistic.

Passed in →	First attempt	Second attempt
Men	32	28
Women	8	12

Correct answer: 1.07

Explanation and steps

The four chi-square contributions total approximately 1.07.

Question 4 - ENTER ANSWER · PG-53 · EX-Q4

800 people were asked whether they wear glasses for reading with following results.

Age	Wear glasses	Do not wear glasses
≤ 30	310	90
> 30	290	110

Compute the χ^2 square statistic.

Correct answer: 2.66

Explanation and steps

The four chi-square contributions total approximately 2.66.

Question 5 - MCQ · PG-53 · EX-Q5(a)

Out of a sample of 120 persons in a village, 80 were administered a new drug for preventing influenza and out of them 18 were attacked by influenza. Out of those who were not administered the new drug, 10 persons were not attacked by influenza:

Prepare: (a) a two-way table showing frequencies.

A.

	Attacked	Not attacked
Drug administered	18	62
Drug not administered	30	10

B.

	Attacked	Not attacked
Drug administered	18	30
Drug not administered	62	10

C.

	Attacked	Not attacked
Drug administered	62	18
Drug not administered	10	30

D.

	Attacked	Not attacked
Drug administered	18	10
Drug not administered	30	62

Correct answer:

	Attacked	Not attacked
Drug administered	18	62
Drug not administered	30	10

Explanation and steps

The administered group contains 18 attacked and 62 not attacked. The remaining 40 persons contain 30 attacked and 10 not attacked.

Question 6 - ENTER ANSWER · PG-53 · EX-Q5(b)

Out of a sample of 120 persons in a village, 80 were administered a new drug for preventing influenza and out of them 18 were attacked by influenza. Out of those who were not administered the new drug, 10 persons were not attacked by influenza:

(b) Compute the χ^2 square statistic.

Correct answer: 30.625

Explanation and steps

The completed 2×2 table and expected frequencies give $\chi^2 = 30.625$.